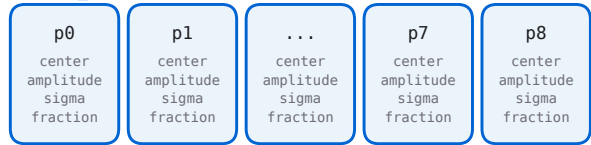


# one model graph

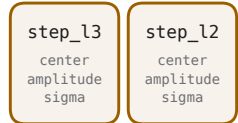
12 declared components carrying 43 parameters, and 2 expression edges; each edge removes one free parameter, leaving 41

## what the analyst declares

components, and the parameter names each one introduces  
pseudo\_voigt (9)



## arctan\_step (2)



## expression edges (2)

$\text{step\_l2.amplitude} = \text{step\_l3.amplitude} / 2$

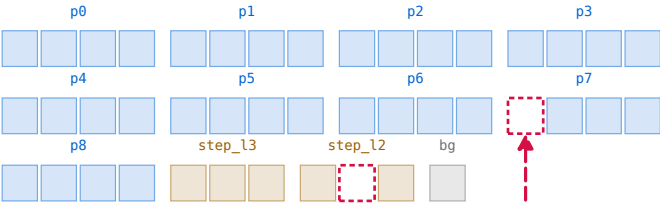
added by scenario B  
a fixed 2:1 intensity ratio between the two absorption edges

$\text{p7.center} = \text{p1.center} + \text{step\_l2.center} - \text{step\_l3.center}$

added by scenario C  
a fixed energy gap between the two main peaks

## what the fit is left to determine

one cell per free parameter, in the order the components declare them



free parameters, scenario by scenario

|   |              |    |    |
|---|--------------|----|----|
| A | free         | 43 |    |
| B | step 2:1     | 42 | -1 |
| C | + spin-orbit | 41 | -1 |

 an open cell is a free parameter an expression edge removed